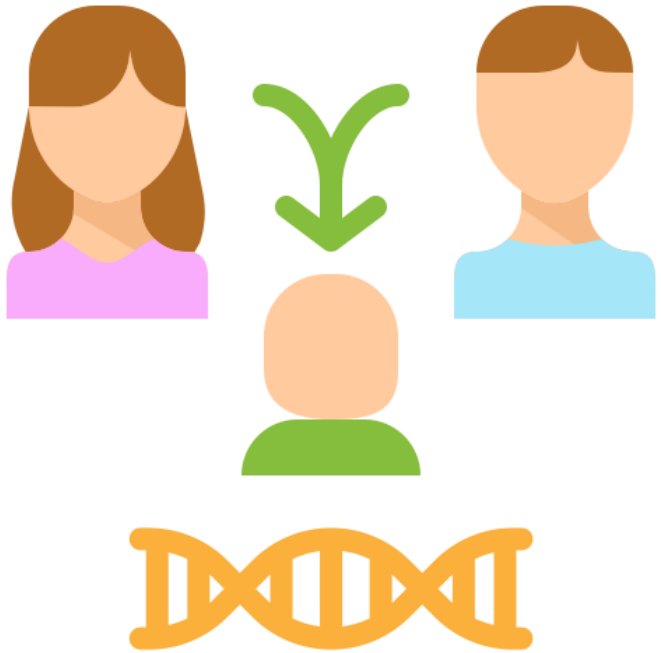
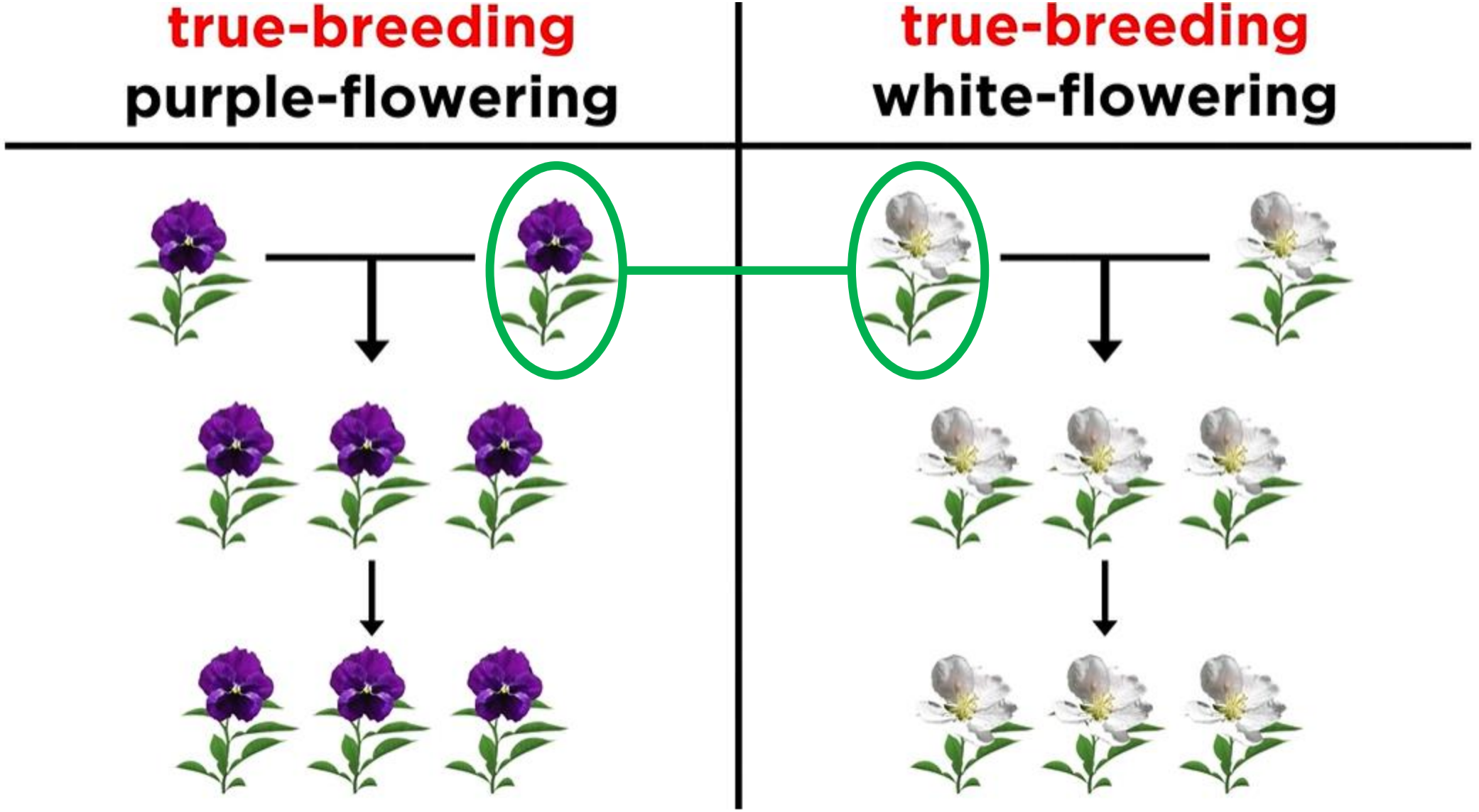
Genetics and Heredity – 2

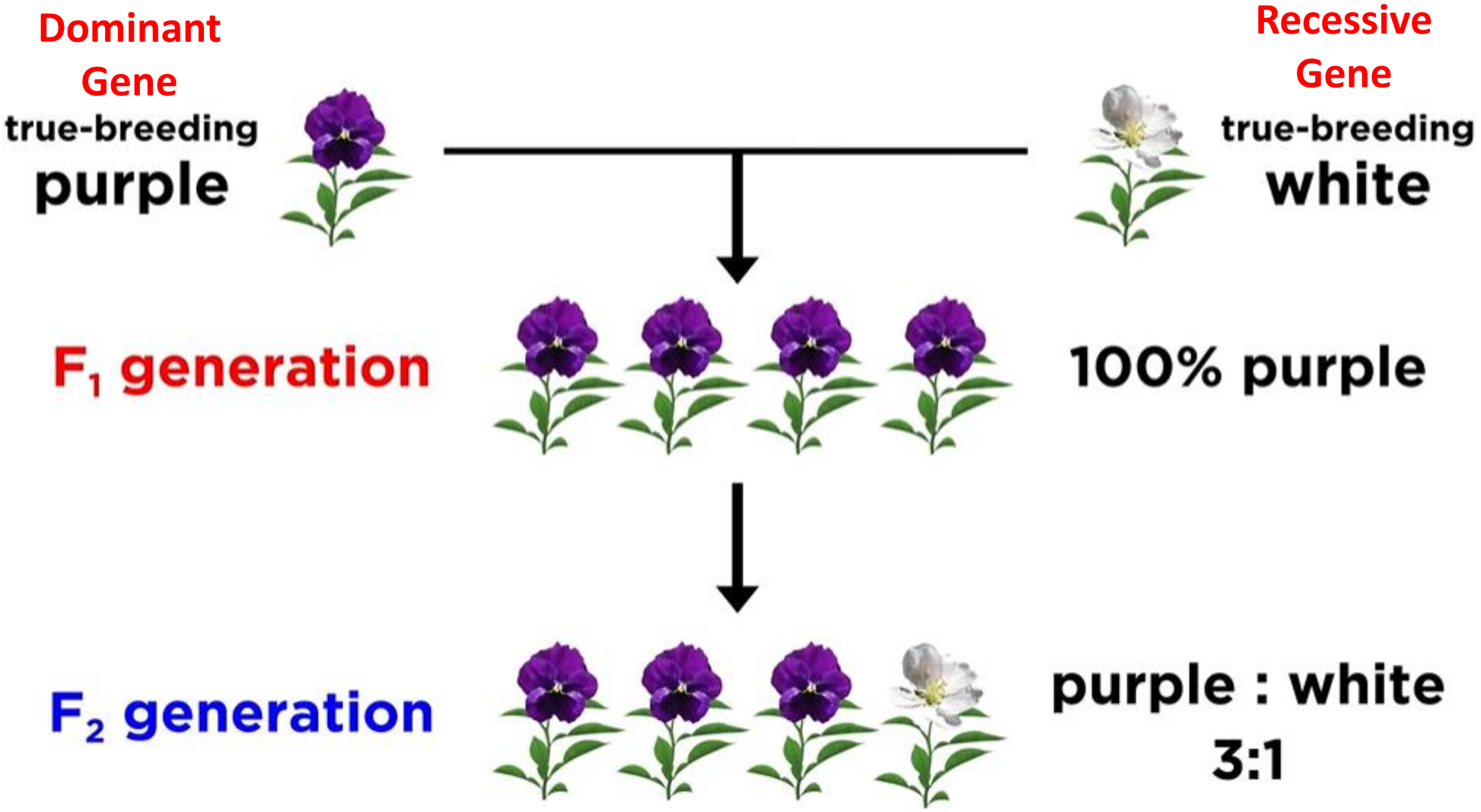
Lecture 7



Mendelian Genetics



Mendelian Genetics



Information for white-flowering plant was not lost in the F1 generation

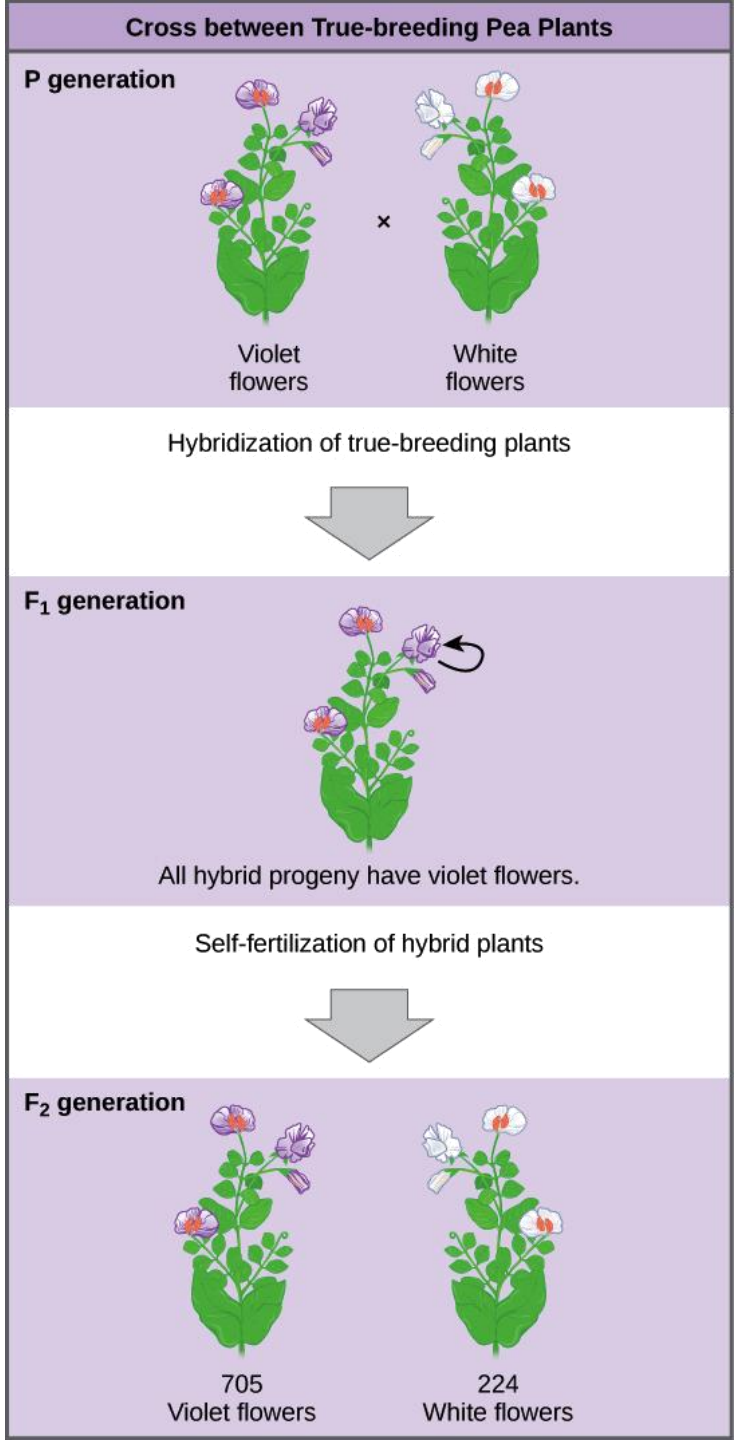
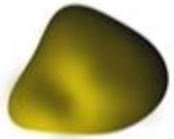


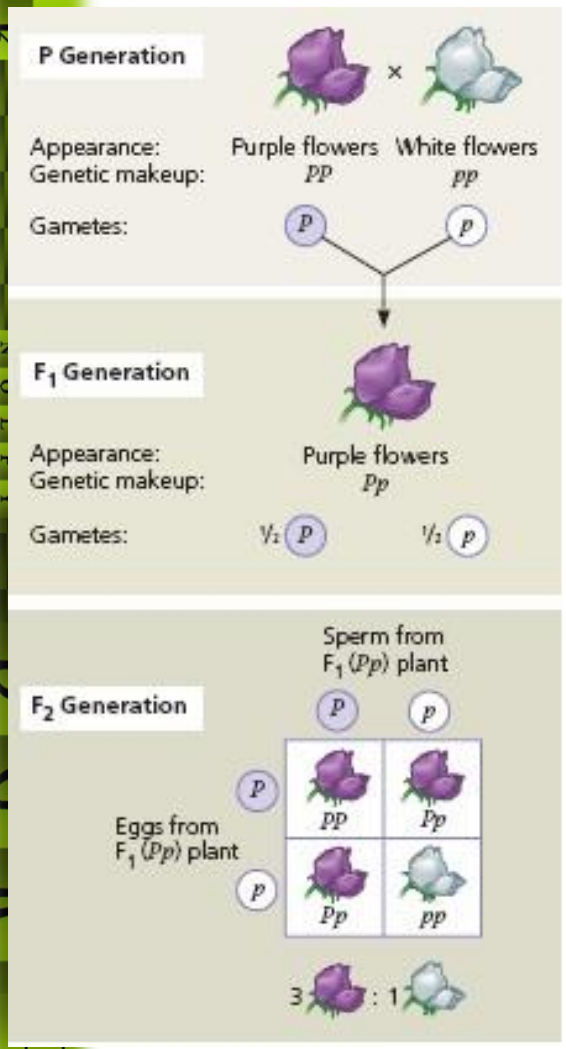
Image courtesy: Principles of Biology by Robert Bear et al., OpenStax

- This 3:1 ratio was consistent across other six characteristics in F2 generation.
- One of the two traits would disappear in F1 to reappear in F2 at 3:1.

Mendelian Genetics

dominant	recessive	F2 phenotype
purple flower 	white flower 	3:1
smooth seeds 	wrinkled seeds 	3:1
yellow seeds 	green seeds 	3:1
green pods 	yellow pods 	3:1

Monohybrid Cross



Character	Dominant Trait	×	Recessive Trait	F_2 Generation Dominant: Recessive	Ratio
Flower color	Purple	×	White	705:224	3.15:1
Flower position	Axial	×	Terminal	651:207	3.14:1
Seed color	Yellow	×	Green	6,022:2,001	3.01:1
Seed shape	Round	×	Wrinkled	5,474:1,850	2.96:1
Pod shape	Inflated	×	Constricted	882:299	2.95:1
Pod color	Green	×	Yellow	428:152	2.82:1
Stem length	Tall	×	Dwarf	787:277	2.84:1

Observations:

Heritable factor for white flowers did not disappear in the F1 plants, but was somehow hidden and reappear in the F2 generation.

Purple flower is a *dominant* trait (*P*)
White flowers is recessive trait (*p*)

Appearance of traits: Phenotype.
Genetic makeup of trait: Genotype.

Organism with two different traits: Heterozygous (Pp).
Organism with identical traits: Homozygous (PP, pp)

genotype	phenotype
PP and Pp	purple
pp	white

In F2 generation
Phenotypic ratio is approximately 3:1
Genotypic ratio is 1:2:1.

Mendel's Explanation:

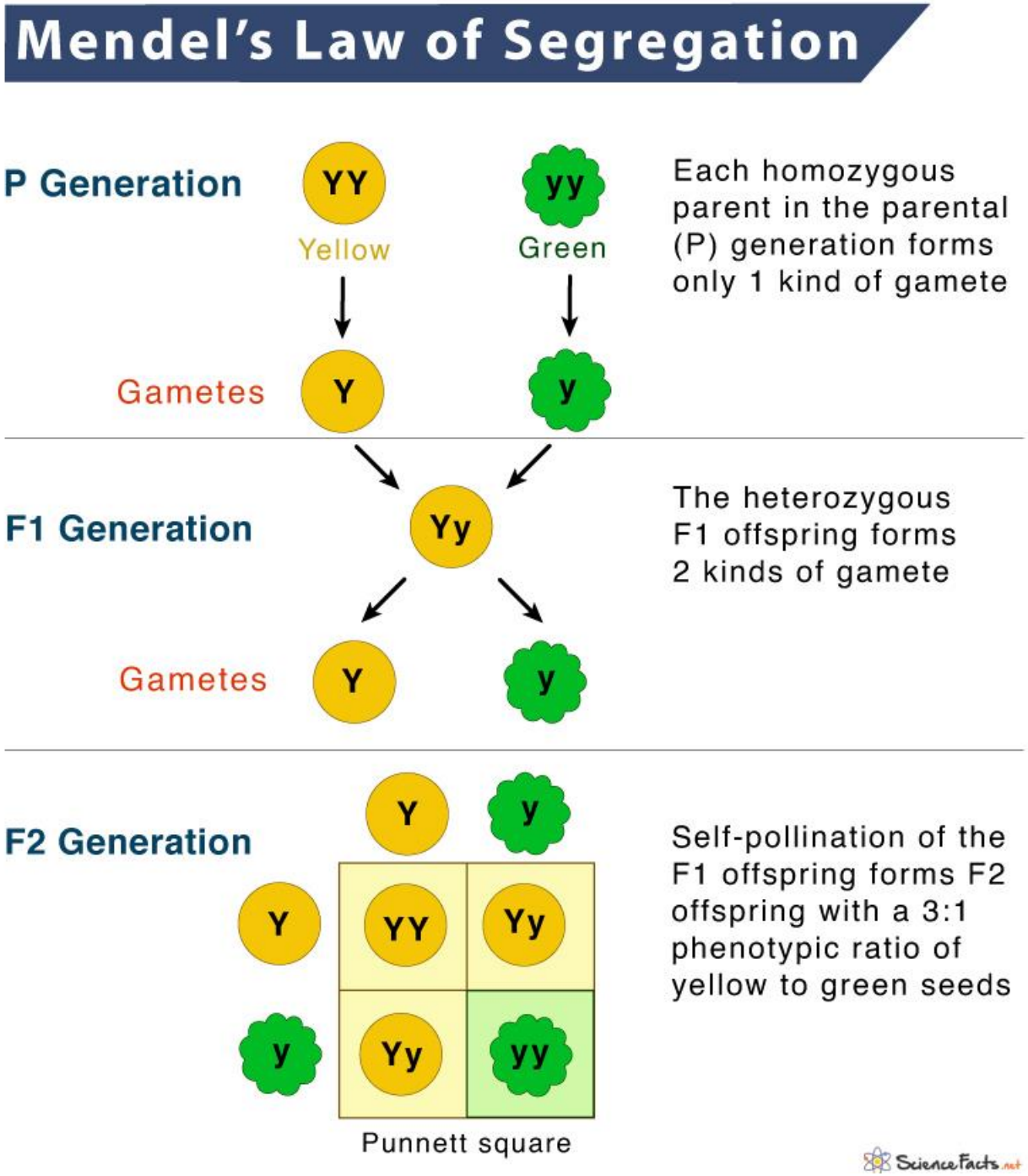
- Alternative versions of genes account for variations of inherited characters. These alternative versions of a gene are called Alleles.
- For each character, an organism inherits two copies of a gene, one from each parent.
- If the two alleles at a locus differ, then one, the dominant allele, determines appearance. The other, the recessive allele, has no noticeable effect.

1. Law of Dominance

- Only one of the two gene copies present in an organism is distributed to each gamete (egg or sperm cell) that it makes, and the allocation of the gene copies is random.

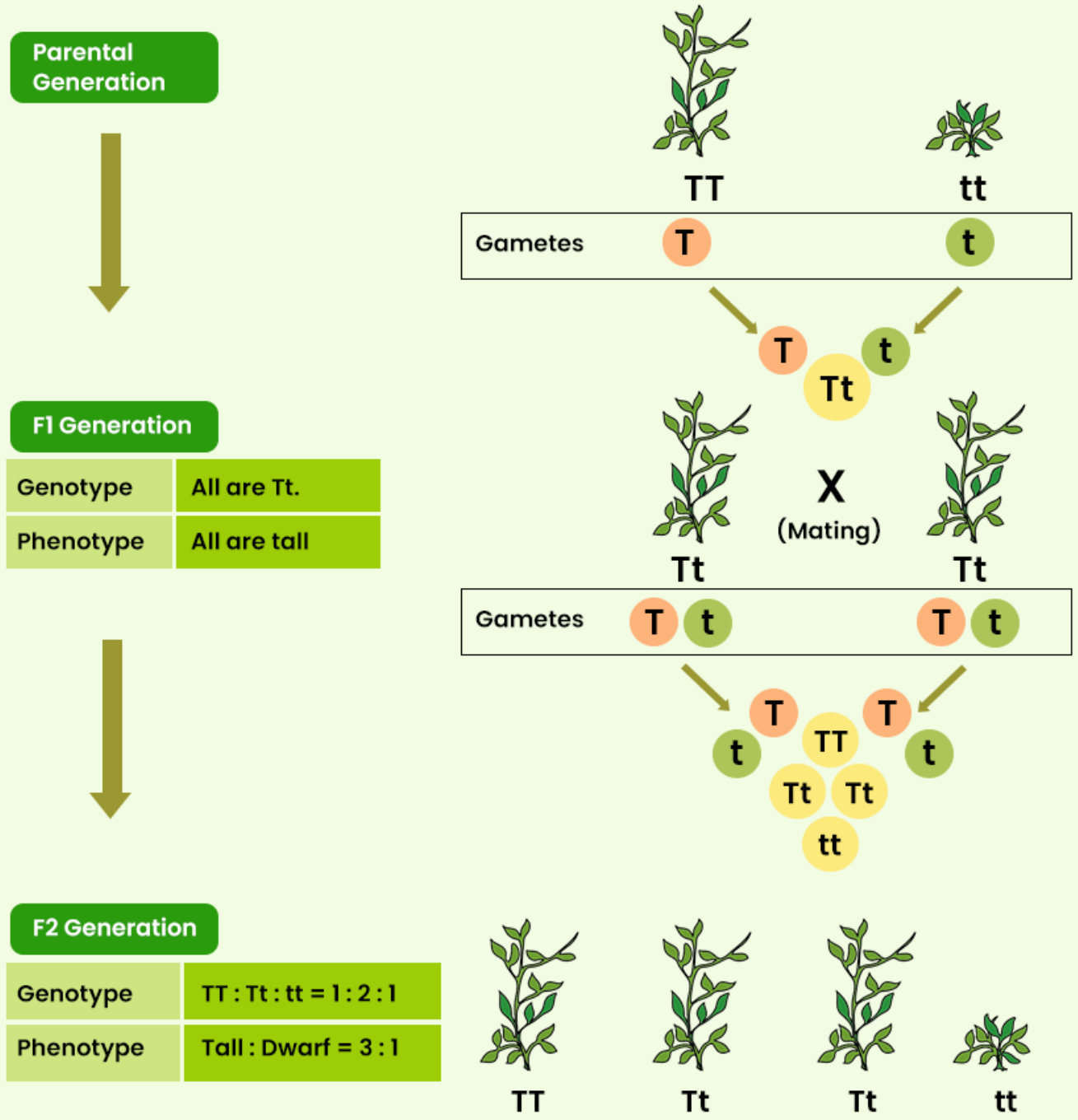
2. The Law of Segregation

The Law of Segregation



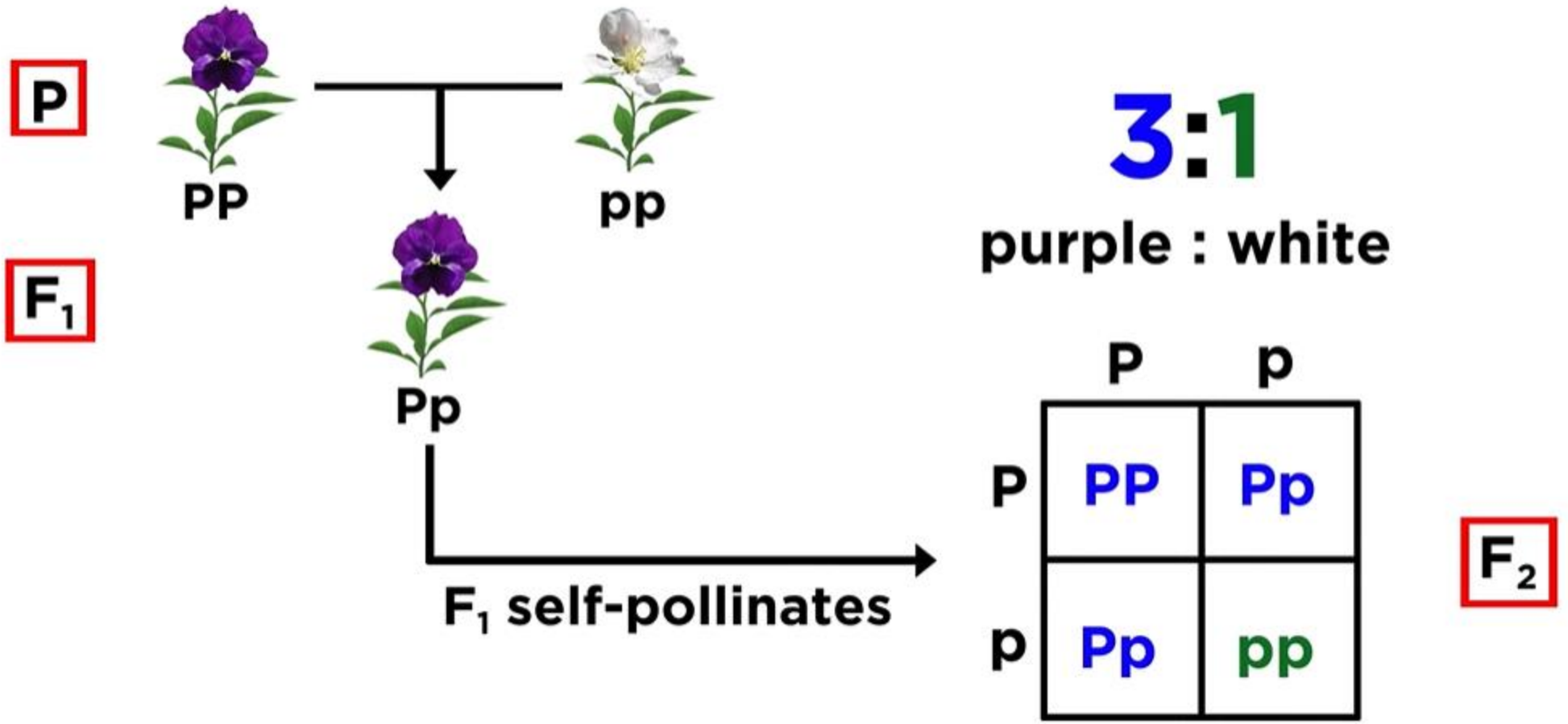
https://www.geeksforgeeks.org/biology/law-of-segregation-law-of-dominance/

The Law of dominance



https://www.geeksforgeeks.org/biology/law-of-segregation-law-of-dominance/

Punnett Squares



Dihybrid Cross

Matings that involve parents that differ in two genes (two independent traits)

For example, Seed color:

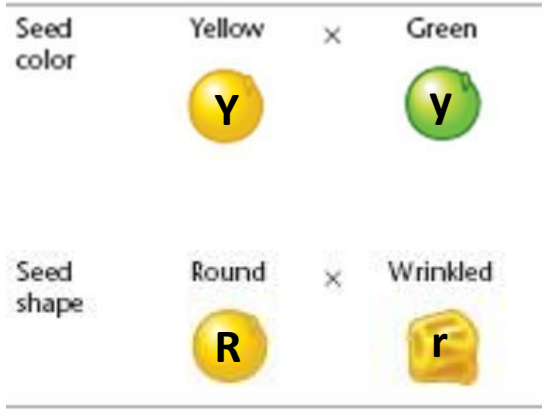
Y = Yellow (dominant)

y = Green (recessive)

And seed shape:

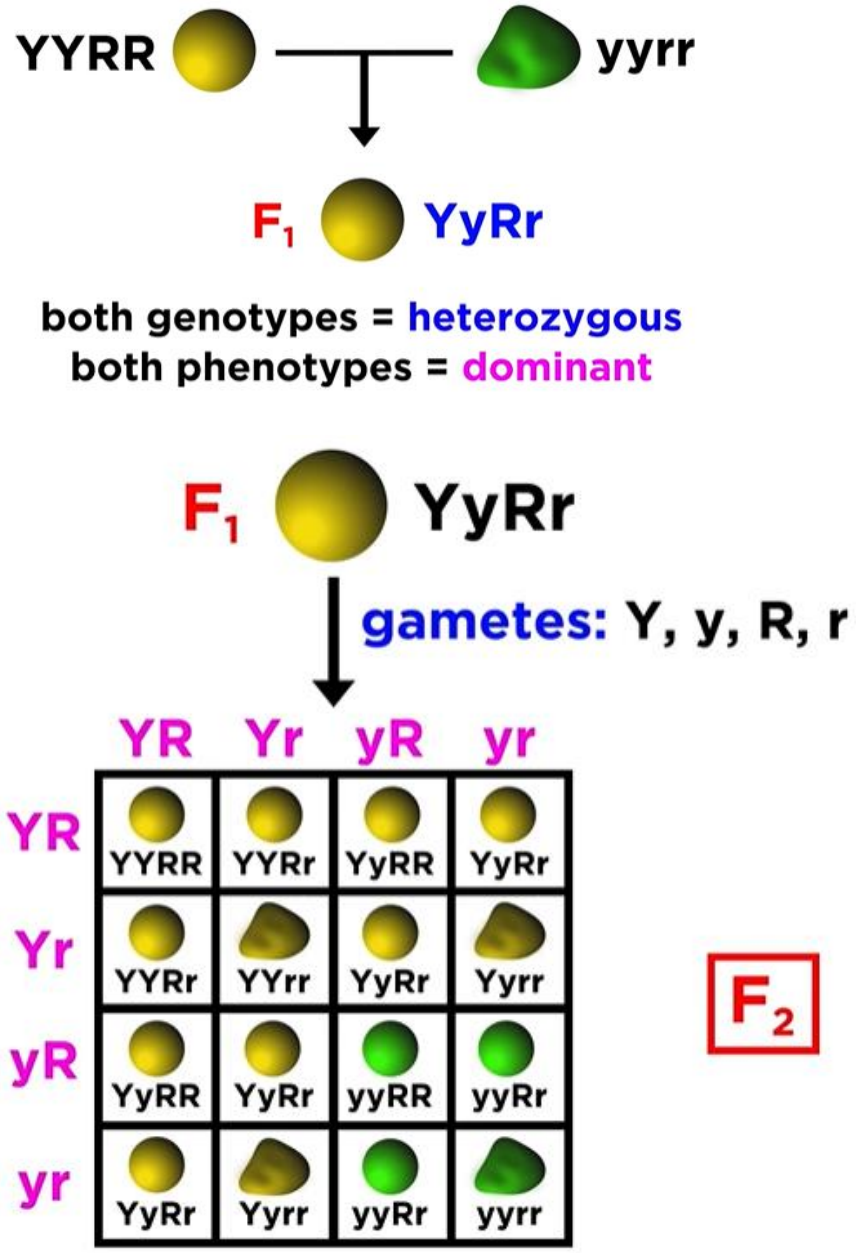
R= Round (dominant)

r= Wrinkled (recessive)



This means that at the time of gamete formation, the two genes segregate independently of each other as well as of other traits.

3. Law of independent assortment



Dihybrid Cross

Matings that involve parents that differ in two genes (two independent traits)

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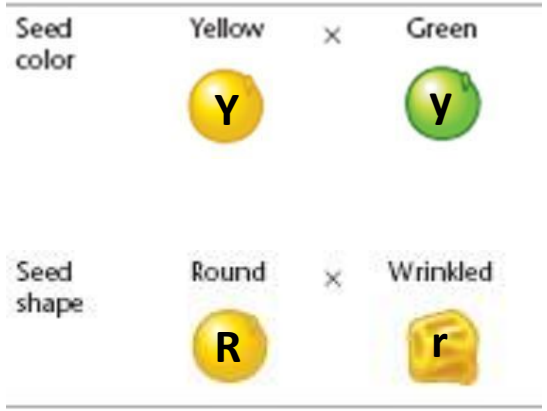
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3. Law of independent assortment

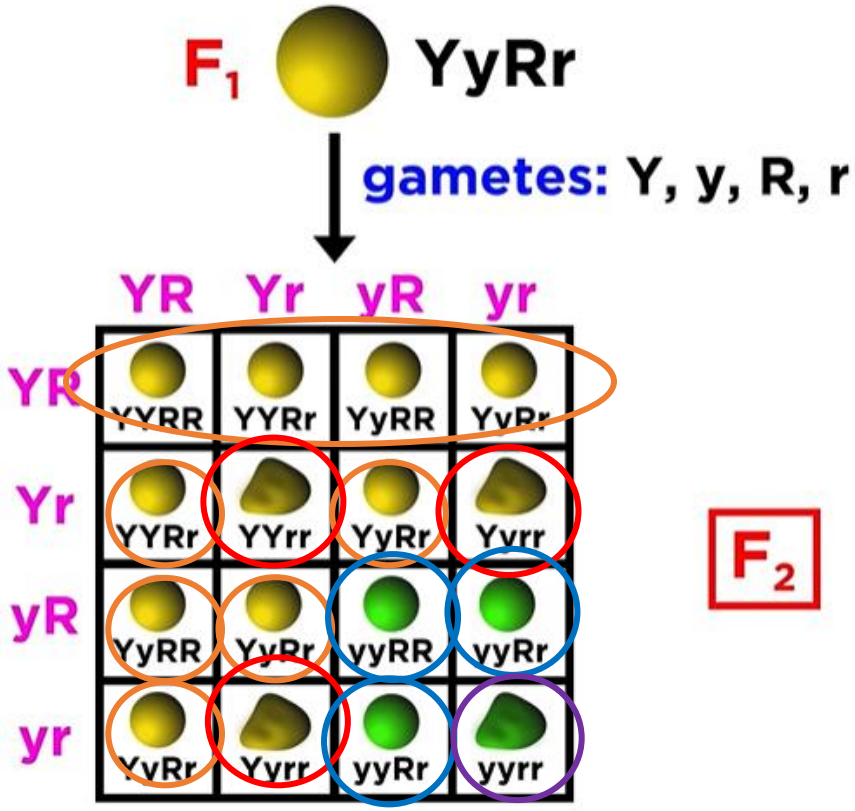
Phenotypic ratio → 9:3:3:1

9 → Both traits dominant

3 → Seed color dominant

3 → Seed shape dominant

1 → both traits recessive



Law of Independent Assortment

Different gene pairs can be packaged into gametes independently.

For genes that assort independently, the allele inherited at one gene does not determine the allele inherited at another.

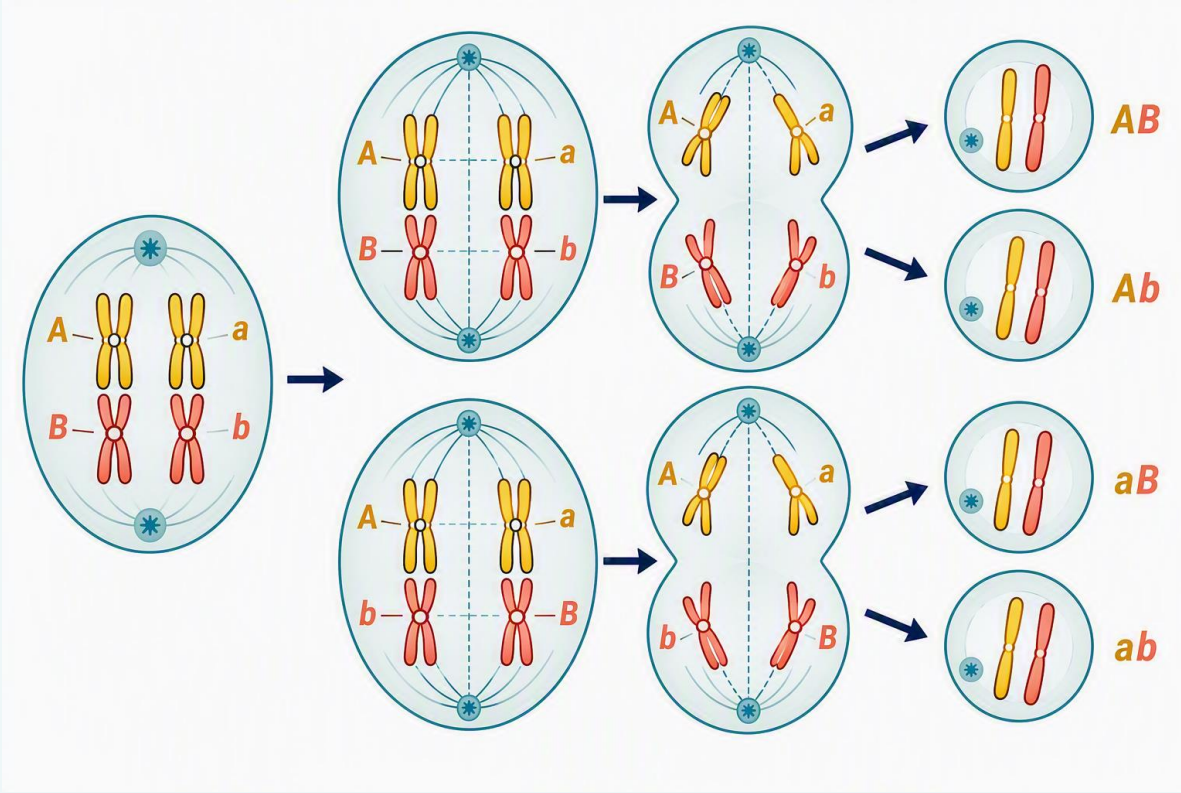
AaBb → AB, Ab, aB, ab

4 equally likely gamete types

Key condition: genes must be unlinked or far apart.

Random orientation at metaphase I

Each homologous pair aligns independently of the other pair.



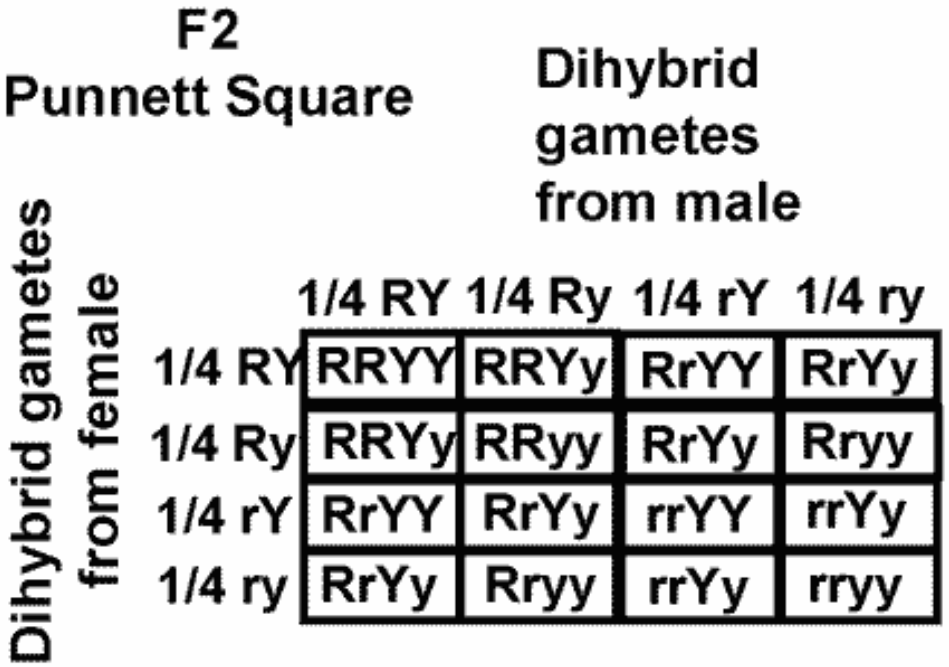
Punnett Squares

Used to predict possible offspring genotypes.

In dihybrid cross, four diallelic combinations of gamete occur with an equal probability of 1/4.

Given four possible gamete types in each parent, 4 x 4 = 16 possible F2 combinations.

The probability of any particular dihybrid type is 1/16.



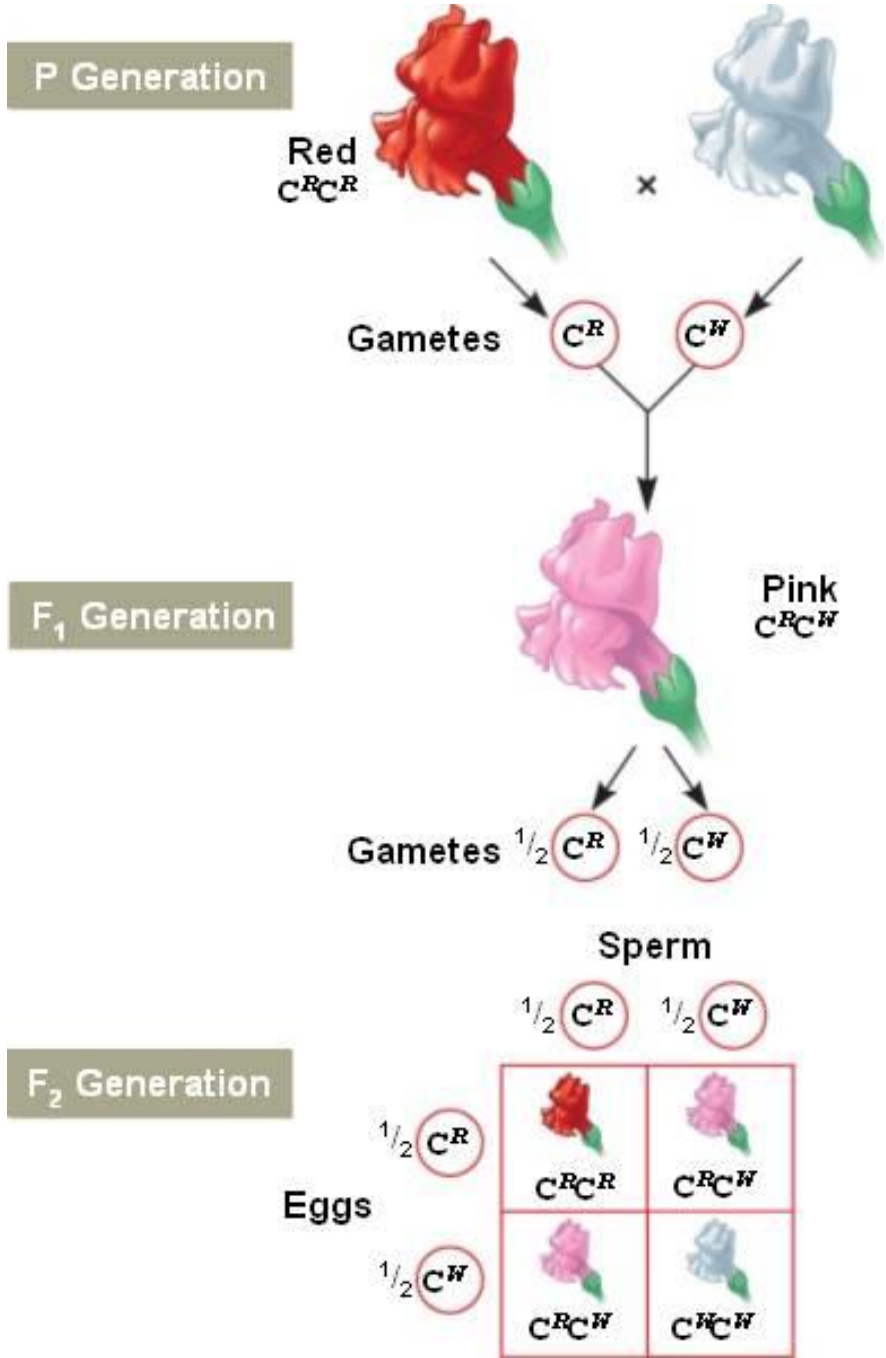
Phenotypic ratio → 9:3:3:1

To predict the genotypic ratios of dihybrid cross:
1 YYRR + 2 YYRr + 1 YYrr + 2YyRR + 4YyRr + 2 yyRR + 1yyRR + 2yyRr + 1yyrr
Genotypic ratio: 1:2:1:2:4:2:1:2:1

Incomplete Dominance

For some genes, neither allele is completely dominant, and the F1 hybrids have a phenotype somewhere between those of the two parental varieties.
This phenomenon is called **incomplete dominance**.

- A condition where all three genotypes are expressed.
- Phenotypic level: contradicts Mendel's conclusions.
- Genotypic level: consistent with Mendel's laws.

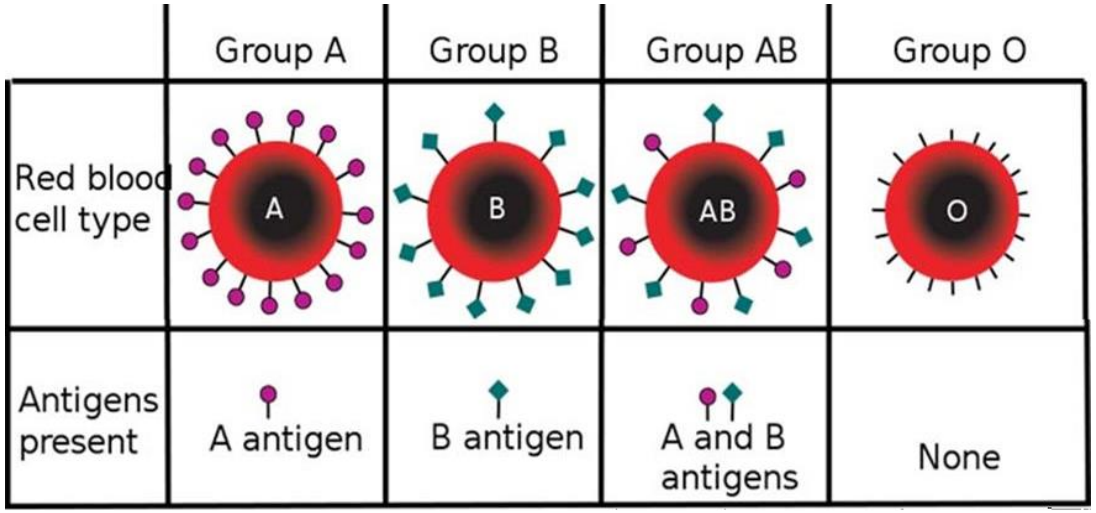


https://bio1152.nicerweb.com/Locked/media/ch14/?C=M;O=A

Co-dominance : Blood Group

Codominance is a type of genetic inheritance where two different versions (alleles) of a gene are fully and equally expressed in the phenotype of an organism.

Blood type AB is an example of co-dominance



Phenotype	Genotype
Type A	I^AI^A or I^Ai
Type B	I^BI^B or I^Bi
Type AB	I^AI^B
Type O	ii

Co-dominance : Blood Group

"A" gives to A,
receives from A and O

"B" gives to B,
receives from B and O

"AB" gives to AB,
receives from A, B, and O

"O" gives to all blood
types, receives only
from O

Determination of ABO Blood Group by Multiple Alleles

Genotype	Phenotype (Blood Group)	Red Blood Cells
I^AI^A or I^Ai	A	
I^BI^B or I^Bi	B	
I^AI^B	AB	
ii	O	

Chromosomes – some facts

- Humans have 46 chromosomes – 23 pairs
- Out of the 23 pairs of chromosomes, one pair contains the sex chromosomes.
- The non-sex chromosomes are referred to as autosomes
  - Traits linked to autosomes are called autosomal traits
- The sex-chromosomes are referred as ‘X’ and ‘Y’ chromosomes
  - Besides sex information, there are almost 900 – 1400 genes in the X chromosome, and about 70 – 140 in the Y chromosome
  - Traits and genes linked to the X chromosome are referred to as “X-linked.”
  - Traits and genes linked to the Y chromosome are referred to as “Y-linked.”

